Treasure Diver 3D – Feasible Features (Strict OpenGL/GLUT)

1. Player Controls

Original Idea:

Move forward/backward/left/right (W, A, S, D)

Move up/down (Q, E)

Rotate camera (mouse/arrow keys)

Feasible Extensions:

Smooth movement using glTranslatef with small increments.

Camera follow mode: gluLookAt(camera_pos, diver_pos, up_vector) so the camera always looks at diver.

Small dodge/slide by temporarily moving diver in one direction using glTranslatef.

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2. Enemies / Hazards

Original Idea:

Sharks (cylinder + cone)

Jellyfish (sphere)

Falling rocks (cylinder)

Feasible Extensions:

Move enemies along fixed paths using glTranslatef.

Random spawn positions with rand() for rocks/enemies.

Simple collision detection using distance check between diver and enemy coordinates.

Color change of enemies when close to the diver (using glColor3f).

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3. Treasures & Scoring

Original Idea:

Treasures: spheres or cubes, +10 points

Rare treasures: +50 points or extra life

Feasible Extensions:

Animate treasures to “bob up and down” using glTranslatef in the idle() function.

Use different colors for normal vs rare treasures.

Display score on screen using draw_text().

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4. Power-Ups

Original Idea:

Oxygen bubble: restores oxygen (sphere)

Speed boost: temporary faster movement

Shield bubble: temporary invincibility (sphere around diver)

Feasible Extensions:

Animate oxygen bubbles to float upward (glTranslatef).

Speed boost: temporarily increase movement increment per frame.

Shield bubble: draw a slightly bigger colored sphere around diver to show invincibility.

Use frame count or idle function for timer duration.

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5. 3D Objects

Original Idea:

Sphere: treasures, jellyfish, oxygen bubbles

Cylinder: sharks, rocks

Cube: treasure chests, obstacles

Cone: shark fins, diver’s flippers

Feasible Extensions:

Combine shapes to form more complex objects (e.g., shark = cylinder + cones).

Rotate objects using glRotatef for simple animation (e.g., shark fins, treasure rotation).

Scale objects with glScalef for variety in size.

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6. Environment & Visuals

Original Idea:

3D grid/floor

Bubbles or floating effects

Feasible Extensions:

Rising bubble effect using glTranslatef and GL_POINTS or small spheres.

Water depth effect: darker/lighter color zones using glColor3f.

Sea floor/grid using GL_QUADS (glBegin(GL_QUADS)) with colored quads.

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7. Camera Features

Original Idea:

Fixed camera or follow diver

Feasible Extensions:

Rotate camera left/right/up/down using arrow keys (specialKeyListener).

Zoom in/out by changing camera_pos z-coordinate.

Camera always looks at diver using gluLookAt.

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8. UI / Feedback

Original Idea:

Display score, oxygen, lives

Feasible Extensions:

Draw text using draw_text() for:

Score

Lives

Oxygen bar (rectangle drawn with GL_QUADS scaled according to remaining oxygen)

Flash text or change color for warnings (low oxygen or enemy near).

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9. Game Mechanics

Original Idea:

Lose life on collision

Game over when oxygen = 0 or lives = 0

Feasible Extensions:

Continuous oxygen drain using idle() function.

Random treasure/enemy respawn after collection.

Simple bounding-box or distance check for collision detection.